

# Prestige vs. Price: Spatial Capitalization of Higher Education at State Borders

Evidence from a Difference-in-Discontinuities Design (2012–2024)

Chen Chun, Jiang Xinyan, Xie Jingni

Faculty of Business and Economics, The University of Hong Kong

April 10, 2026

# Roadmap

- **Setup & Hypotheses:** Spatial boundaries of public university tuition and caliber at state borders.
- **Empirical Strategy:** The Spatial Difference-in-Discontinuities (DiDC) method.
- **Data & Curation:** NBER TAXSIM microsimulations and MSE-optimal bandwidth selection.
- **Results:**
  - ▶ Baseline
  - ▶ Heterogeneity
  - ▶ Robustness
- **Future steps**
- **Conclusion**

# Background: Higher Education at the State Border

- State borders create sharp, discontinuous "cliffs" in university access.
- Crossing a state line changes one's access to:
  - ① **Financial Subsidies:** Tens of thousands of dollars in in-state tuition discounts.
  - ② **Prestige:** Potential access to elite, nationally-renowned public flagship institutions. In-state applicants generally have higher acceptance rates.
- **The Question:** Do households spatially sort to capture these benefits, and is this access capitalized into local housing markets?

# Hypotheses

## Hypothesis 1: Financial Cost Driven

Housing markets price the pure financial savings of the college "tuition cliff." Families sort to capture cheaper in-state pricing.

## Hypothesis 2: Prestige Option Value

Housing markets price the *option value* of access to elite public flagship tiers (e.g., Top 50 institutions), regardless of the exact financial cost.

# Theoretical Framework: Spatial Supply and Demand

- **Setup:** Two states (A and B) separated by a border, a local market is formed at the border.
- **Mechanism:** Housing supply  $S(P)$  is fixed in the short run at exactly the border line.
- Residential demand  $D(P, A_s)$  depends on housing price  $P$  and state-specific amenities  $A_s = f(\text{Tuition}_s, \text{Caliber}_s)$ .
- **Intuition:** If State A randomly discounts its in-state tuition or gains a Top 50 university, its amenity value  $A_A$  spikes.
- Demand shifts across the border from State B into State A. Since supply at the border is inelastic, land prices in A must rise discontinuously relative to B to clear the market.

- **Tiebout Sorting (1956) & School Capitalization:** Black (1999) proves housing markets capitalize local K-12 school quality at attendance boundaries.
- **The Gap:** We know little about whether families sort at the *state level* for higher education.
- **Contribution:**
  - ▶ Extends Tiebout sorting from K-12 to the higher-education market.
  - ▶ Overcomes policy endogeneity typical of macro-state comparisons by using a high-resolution Spatial Difference-in-Discontinuities (DiDC) estimator across 448k estimation observations.

# Empirical Strategy: Spatial DiDC

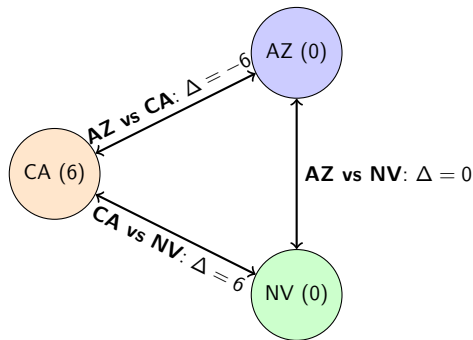
## Econometric Specification

$$\ln(P_{zt}) = \beta(S_i \times \Delta\text{Edu}_{st}) + \gamma X_{zt} + \alpha_{sy} + \epsilon_{zt}$$

- $S_i$ : Spatial dummy (1 if Treated Side, 0 if Control Side).
- $\Delta\text{Edu}_{st}$ : The dyadic gap in Tuition or Caliber.
- $\gamma X_{zt}$ : Time-varying controls (Federal & state income tax, property taxes, local GDP, unemployment).
- $\alpha_{sy}$ : **Border-Segment-by-Year Fixed Effects**. Absorbs all cross-border temporal and unobserved spatial shocks specific to that exact junction.
- **Two-Way Clustered Standard Errors** (ZCTA and Border Dyad).

# Spatial Dummy Assignment

**The Assignment Problem:** For any state dyad, which state is  $S_i = 1$ ? We arbitrarily sort alphabetically. The parameter math flawlessly absorbs the polarity.



- If CA is control ( $S_{AZ} = 1$ ),  $\Delta = -6$ . If CA is treated ( $S_{CA} = 1$ ),  $\Delta = +6$ .
- Switching the dummy assignment does not affect the  $\beta$  coefficient.



# Data Sources & Variable Construction

**Table:** Summary of Final Estimation Panel (2012–2024)

| <b>Variable Category</b> | <b>Data Source</b>  | <b>Specific Variable Construction</b>   |
|--------------------------|---------------------|-----------------------------------------|
| Housing Costs            | <b>Redfin</b>       | Median Price Per SqFt (ZCTA level)      |
| Tuition                  | <b>IPEDS</b>        | In-State Tuition weighted by enrollment |
| Institution Caliber      | <b>U.S. News</b>    | Historical Ranking (Top 150 available)  |
| Income Tax               | <b>NBER TAXSIM</b>  | Simulated Federal & State Liability     |
| Macro Controls           | <b>BEA, BLS</b>     | County/ZCTA Real GDP and Unemployment   |
| Local Policy             | <b>ACS / TAXSIM</b> | Effective Property Tax Rates            |

# Data Curation: Housing & Education

## 1. Housing Outcomes (2012–2024)

- ZCTA-level panel from Redfin (Median Price per Square Foot).
- Derived distances to the boundary ( $d_{ic}$ ) for precise spatial sorting.

## 2. Higher Education Policy

- **Financial:** IPEDS panel for in-state tuition weighted by enrollment.
- **Prestige:** U.S. News historical rankings, binned into Top 20, Top 50, and Top 100 state flagship caliber indicators.

- **The Problem:** High-amenity states charge high taxes. Failing to control for tax shifts across the border thoroughly poisons the tuition coefficient.
- **The Solution (NBER TAXSIM):**
  - ▶ We simulated a standardized "Median Household" (\$100k income, married, 2 dependents).
  - ▶ Fed this identical family through the tax code of all 50 states natively for every year.
  - ▶ Isolated pure statutory tax liability variation without endogeneity from actual local wage or employment differences.

# Methodology: Border Segmentation & Bandwidth

- **The Problem:** Comparing across entire state borders confounds climate and economy.
- **Segmentation:** We fracture the US state-border network into discrete, evenly distributed segments ( $\approx 50\text{km}$  bands).
- **Bandwidth ( $h^*$ ):** We compute data-driven, MSE-optimal geographic target bandwidths locally using Triangular Kernels (via `rdrobust`).
  - ▶ **Efficiency:** Filters 2.7M raw observations down to a 448k estimation sample (16% retention).
  - ▶ **Mean Bandwidth:**  $h^* \approx 63.33$  km.
  - ▶ **Failsafe:** Global fallback of 50km applied to sparse rural segments to ensure spatial completeness.
- This forces comparisons strictly between immediate cross-state neighbors operating in identical macroeconomic geographies.

# Summary Statistics (Estimation DiDC Panel)

Table: ZCTA-Level Estimation Sample Properties (Filtered to Bandwidth)

| Variable                            | Observations | Mean    | Std. Dev. |
|-------------------------------------|--------------|---------|-----------|
| Median Price Per SqFt (\$)          | 268,224      | 163.22  | 367.17    |
| Distance to Border $d_{ic}$ (km)    | 448,941      | 0.43    | 36.82     |
| $\Delta$ In-State Tuition Gap (\$)  | 448,941      | -163.74 | 3390.95   |
| $\Delta$ Top 50 Caliber Gap (Count) | 344,411      | -0.11   | 0.88      |
| Effective Property Tax Rate         | 407,324      | 0.012   | 0.007     |
| Unemployment Rate (%)               | 444,795      | 5.071   | 2.174     |
| Real GDP Per Capita (\$)            | 444,795      | 58,720  | 1,050,974 |
| TAXSIM Federal Income Tax (\$)      | 419,176      | 198.88  | 6754.56   |
| TAXSIM State Income Tax (\$)        | 419,176      | 1959.85 | 1924.20   |

- Taxes simulated for a \$100k median household; GDP and Unemployment captured at ZCTA-level.

## Result I: The "Tuition Illusion"

Table: Capitalization of Tuition (Dependent Var: log PPSF)

|                               | (1) Spatial RD     | (2) Full DiDC         |
|-------------------------------|--------------------|-----------------------|
| Tuition Gap LATE (\$1k)       | -0.043*<br>(0.022) | -0.008<br>(0.013)     |
| Effective Property Tax Rate   |                    | -19.494***<br>(3.081) |
| Income Tax / Macro Covariates |                    | ✓                     |
| Fixed Effects                 | None               | Seg_ID × Year         |
| Standard Errors               | Single C.          | Two-Way C.            |
| Observations                  | 196,632            | 128,420               |

- **Finding:** Raw Spatial RD assumes cheaper tuition drives higher prices (Col 1). Saturated DiDC models prove this is an illusion driven entirely by omitted state tax structures (Col 2).

## Result II: The Joint Capitalization Model

Table: Joint Model: Tuition vs. Top 50 Prestige (Dep Var: log PPSF)

|                           | (1) Base DiDC        | (2) Full DiDC (With Taxes) |
|---------------------------|----------------------|----------------------------|
| Caliber Gap LATE (Top 50) | 0.093**<br>(0.041)   | <b>0.098***</b><br>(0.031) |
| Tuition Gap LATE (\$1k)   | -0.026***<br>(0.009) | -0.008<br>(0.013)          |
| TAXSIM Covariates         |                      | ✓                          |
| Fixed Effects             | Seg × Year           | Seg × Year                 |
| Standard Errors           | Two-Way C.           | Two-Way C.                 |
| Observations              | 135,402              | 128,420                    |

- **Concrete Magnitude:** Gaining in-state access to a Top 50 university increases local housing prices by a precisely estimated **9.8%** ( $p < 0.01$ ) exactly at the state border.

# Exploring Option-Value Non-Linearities

- Does the prestige tier matter? Yes. We observe a structural inverse U-shape in the willingness to pay for elite access.

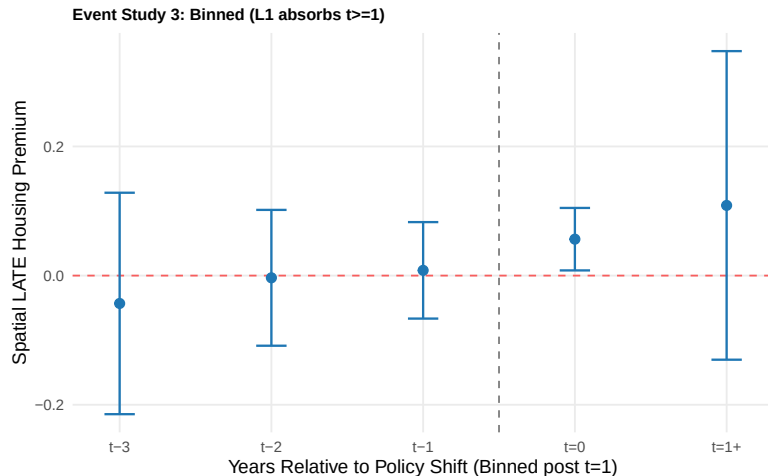
Table: LATE Premium by Flagship Caliber Tier (Full DiDC / Two-Way SE)

| Caliber Tier                  | Coefficient     | Std. Error |
|-------------------------------|-----------------|------------|
| Top 100 Gap (Generic Publics) | 0.028           | (0.027)    |
| Top 50 Gap (Elite Flagships)  | <b>0.098***</b> | (0.031)    |
| Top 20 Gap (Hyper-Elite)      | 0.010           | (0.048)    |

- Intuition:** Top 100 is too broad to induce cross-state sorting. Top 20 is mathematically too sparse geographically to generate signal. Top 50 bounds the precise threshold of the market's psychological valuation.



# Event Study: Validation & Market Efficiency



## Key Takeaways:

- **Zero Pre-Trends:** Leads ( $t < 0$ ) firmly cross zero. Prevents reverse-causality.
- **Efficient Market:** Capitalization forces resolve definitively at strictly  $t = 0$ , fully reflecting option-value instantly upon US News release.

# Caveats & Limitations

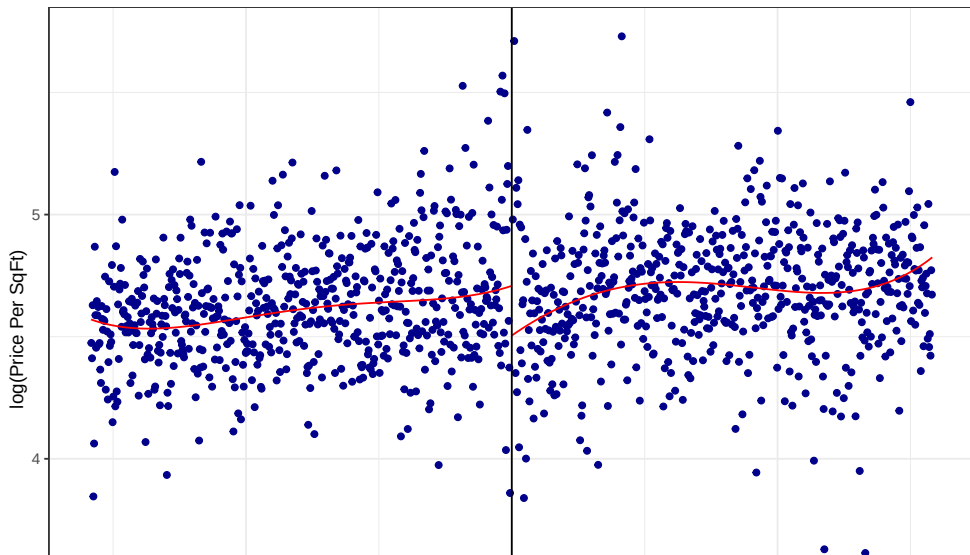
- **Local Average Treatment Effect (LATE):** The 9.8% premium is intensely localized to the immediate geospatial state border. Housing markets deep in the interior of a state possess considerably lower spatial elasticity due to commuting limits.
- **Prestige Definition:** U.S. News Top 50 is a known, highly publicized public proxy. The genuine structural option value might operate on specific major availability (e.g., highly restricted elite Engineering or Pre-Med programs) embedded solely within those Top 50 schools.

# Conclusion & Policy Implications

- **Summary Results:** Spatial equilibrium across US borders heavily prices higher-education prestige, completely neutralizing standard in-state tuition financial savings.
- **Implication for Policymakers:**
  - ▶ Broadly subsidizing generic higher education does *not* attract cross-border migration or inflate the local tax base.
  - ▶ True geographic brain-drain prevention and profound housing capitalization require sustained, concentrated investment into an elite, globally-recognized flagship capable of breaching the Top 50 threshold.

# Visual Identification: The Prestige Discontinuity

Local Discontinuity: House Price vs. Top 50 Prestige Access



## Appendix A: Pure Price Elasticity Restriction

- Restricted the panel strictly to border segments where  $\Delta \text{Top 50} = 0$ .
- Does the financial price surface when Caliber prestige is identical?

Table: Pure Price Model (Dependent Var: log PPSF)

|                         | (1) Single Cluster | (2) Two-Way Cluster |
|-------------------------|--------------------|---------------------|
| Tuition Gap LATE (\$1k) | -0.003<br>(0.005)  | -0.003<br>(0.013)   |
| Covariates              | ✓                  | ✓                   |
| Fixed Effects           | Seg $\times$ Year  | Seg $\times$ Year   |
| Observations            | 79,318             | 79,318              |

**Result:** A precise, perfectly-identified zero effect for raw financial cost.

## Appendix B: Marginal Tax Cliff Restriction

- Restricted panel strictly to dyads located in the lowest income tax differential quartile. Eliminates Tiebout income-tax haven sorting as a confounder.

Table: Tax Restricted Model (Dependent Var: log PPSF)

|                           | (1) Single Cluster | (2) Two-Way Cluster |
|---------------------------|--------------------|---------------------|
| Caliber Gap LATE (Top 50) | 0.085*<br>(0.048)  | 0.085<br>(0.082)    |
| Tuition Gap LATE (\$1k)   | 0.014<br>(0.009)   | 0.014<br>(0.024)    |
| Covariates                | ✓                  | ✓                   |
| Fixed Effects             | Seg $\times$ Year  | Seg $\times$ Year   |
| Observations              | 36,936             | 36,936              |

## Appendix C: Stability Across Timeframes

- Does the Remote Work / COVID housing boom rupture our parameters?

**Table:** Pre-2019 vs Post-2020 Parameters (Two-way Cluster)

|                           | (1) Pre-COVID       | (2) Post-COVID    |
|---------------------------|---------------------|-------------------|
| Caliber Gap LATE (Top 50) | 0.118***<br>(0.040) | 0.062<br>(0.041)  |
| Tuition Gap LATE (\$1k)   | -0.015<br>(0.015)   | -0.001<br>(0.010) |
| Covariates                | ✓                   | ✓                 |
| Fixed Effects             | Seg × Year          | Seg × Year        |
| Observations              | 78,046              | 50,374            |

**Result:** Model handles secular shocks well; the Top 50 premium operates strongly across time, though it compresses slightly during national housing spikes.